

AI Portfolio

AI Architecture Advisor

8-Agent Pipeline — Discovery to CFO Narrative

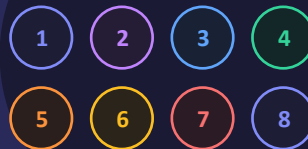
Demo Mode

Claude API

OpenAI API

danvzla.github.io/ai-architecture-advisor

Daniel Mazzini · Principal Architect & Senior TPM



Every enterprise AI engagement starts the same way and takes 5-7 days before any work begins

Day 1-2	Discovery & problem framing	Senior Architect
Day 2-3	Reference architecture design	Principal Architect
Day 3-4	Architecture Decision Records	Principal Architect
Day 4-5	Risk register & governance controls	Risk Architect
Day 5-6	30/60/90 day delivery roadmap	Senior TPM
Day 6-7	ROI model & CFO executive narrative	Value Engineer

5-7 days. Each engagement starts from scratch. ADRs are almost never written until an audit forces it. The ROI case gets assembled after the architecture decision — advocacy, not analysis.

Seven enterprise-grade deliverables from one discovery session

1

Reference Architecture

Layered target-state diagram with specific technologies per layer

2

Architecture Decision Records

Context · decision · rationale · tradeoffs · alternatives rejected

3

Risk Register & Governance

Risks ranked by severity, category, likelihood, mitigation, owner

4

30/60/90 Day Roadmap

Phased milestones, deliverables, dependencies, go-live criteria

5

TCO / ROI / NPV Model

Full financial model with sensitivity analysis at 65/100/125%

6

CFO Executive Narrative

Investment justification, urgency driver, recommended decision

7

AI Readiness Scorecard

6-dimension maturity score with gap analysis and legend

Each agent receives all prior agents' outputs as context

01 Discovery

problemStatement · rootCauses · stakeholderMap

02 Outcome

outcomeHierarchy · successCriteria

03 Capability

recommendedPattern · capabilityMap · agentDesign

04 Architecture

targetArchitectureLayers · adrs · integrationPatterns
← prior[discovery]+prior[capability]

05 Data & Platform

ragPipelineDesign · platformArchitecture
← prior[architecture]

06 Governance

riskRegister · governanceControls
← prior[architecture]+prior[data]

07 Roadmap

roadmap · goLiveCriteria · teamStructure
← prior[governance].riskRegister

08 Value Engineering

cfoNarrative · investmentJustification
← prior[ALL] + financials

The financial math is never AI-generated — only AI-narrated

Step 7 wizard inputs (volume, labor cost, error rate...)

financial.js
Pure deterministic function

Agent 08 receives the calculated numbers and writes the CFO narrative around them

Why this matters:

- ◆ Most AI tools let the LLM calculate ROI directly — which means it can hallucinate numbers
- ◆ This tool calculates TCO, ROI, NPV, payback, and cost of inaction with deterministic math first
- ◆ Agent 08 never does arithmetic — it only writes the narrative around numbers that are already correct
- ◆ The CFO narrative is persuasive writing on top of verified math, not a number the AI invented

Zero setup to live AI generation — one click apart



No API key required

Demo Mode

Select Demo → choose a scenario → click Quick Demo. Full blueprint loads in ~8 seconds using a pre-built Telco NOC scenario. All 8 agents animate, all 7 tabs populate.

Best for: Always ready — works offline, works from the desktop



claude-haiku-4-5

Claude API

Switch to Claude → enter sk-ant-... key → run. Each of the 8 agents makes an independent API call, results chain forward. Full reasoning grounded in your discovery inputs.



gpt-4o-mini

OpenAI API

Switch to OpenAI → enter sk-... key → run. Identical 8-agent pipeline, same 7 output tabs. Provider is swapped by one line in api-client.js.

Mirrors how a senior architect actually runs discovery

1 Profile

Industry · function · company size · AI maturity (1-6) · urgency

2 Problems

Problems to solve · pain level · stakeholders · description

3 Outcomes

Target outcomes · success metrics · impact timeframe · adoption target

4 Capabilities

Required AI capabilities · autonomy level · UX · multi-agent need

5 Data

Knowledge sources · data quality · deployment model · integrations

6 Governance

Industry risk · governance maturity · auditability · internal capability

7 Value

Annual volume · effort · labor cost · error rate · implementation cost

⚡ Step 7 inputs drive the financial model — labor savings, rework reduction, delay avoidance, TCO, ROI, NPV, payback, cost of inaction

Three layers · browser-native · build step for deployment

PRESENTATION

- ◆ `index.html` — mode selector, scenario dropdown, wizard shell
- ◆ `assets/css/styles.css` — full design system, dark theme
- ◆ Wizard overlay · agent pipeline animation · 7-tab results
- ◆ Browser-native · GitHub Pages · zero infra cost

Built via `node build.js`
`dist/index.html` self-contained

ORCHESTRATION

- ◆ `assets/js/app.js` — mode mgmt, pipeline loop, export
- ◆ `AGENT_REGISTRY[]` — ordered agent array (10 lines)
- ◆ Prior results accumulated, passed to each agent
- ◆ `FinancialModel.calculate()` runs before Agent 08

Pure JavaScript
No bundler dependencies

AGENT + API

- ◆ `agent-01` → `agent-08` (8 individual files)
- ◆ `api-client.js` — Claude ↔ OpenAI provider swap
- ◆ `demo-blueprint.js` — pre-built output, no API key
- ◆ `scoring.js` + `financial.js` — pure math functions

1-line provider swap
in `app.js`

One file per agent · 5-attempt JSON repair chain

Agent file interface (identical for all 8)

```
window.Agent04Architecture = {  
  id: 'architecture',  
  name: 'Architecture Agent',  
  mission: 'Target architecture · ADRs',  
  tokens: 4500,  
  
  buildPrompt(answers, prior) {  
    // wizard answers + prior[discovery]  
    // + prior[capability]  
  },  
  
  async run(apiKey, answers, prior) {  
    const raw = await ApiClient.call(  
      apiKey, this.buildPrompt(answers, prior),  
      this.tokens);  
    return ApiClient.parseJSON(raw);  
  },  
};
```

JSON Repair Chain

5 attempts — one more than the other AI Portfolio tools
(handles whitespace-run collapsing too):

- 1 Direct `JSON.parse()`
- 2 Strip control chars
- 3 Collapse literal newlines
- 4 Collapse whitespace runs
- 5 Close unclosed brackets

Maintenance rule: adding agent 9 = one new file + one line in `agents.js`

Broadest vertical coverage across the AI Portfolio

01

Telco Network AIOps Platform

Telecommunications

02

Financial Services GenAI Assistant

Financial Services

03

Healthcare Clinical Decision Support

Healthcare

04

Government Document Intelligence

Public Sector

05

Enterprise IT Operations (ServiceNow)

Technology

06

Retail Customer Experience AI

Retail

07

Manufacturing Predictive Maintenance

Manufacturing

08

Cybersecurity Threat Intelligence

Technology

09

Insurance Claims Processing AI

Financial Services

10

Energy Grid Optimization AI

Energy

11

Legal Contract Intelligence

Professional Services

12

Supply Chain Intelligence

Manufacturing

13

HR Talent Intelligence Platform

Technology

14

Pharma Clinical Trials AI

Healthcare

15

Private Cloud AI Infrastructure

Technology

Telco Network AIOps Platform — Scenario 01

Financial Model

Annual gross benefit	\$845K
3-year ROI	87%
NPV @ 8% discount	\$1.1M
Payback period	14 months
Cost of inaction (3yr)	\$2.6M

Architecture Pattern Selected

RAG + Agentic AI
with Human-in-the-Loop
and Zero Trust Enterprise Integration

Sample ADR-002

Decision: Primary: Azure OpenAI gpt-4o. Fallback: Anthropic Claude claude-haiku-4-5 via Azure Marketplace.

Rationale: Azure OpenAI meets data residency compliance. Dual-provider fallback ensures 99.9% AI availability.

Alternatives rejected: OpenAI direct API (data leaves Azure boundary). Single provider (unacceptable availability risk for 24/7 NOC).

What this demonstrates — and what it replaces

~8s

Demo Mode

Full 8-agent blueprint, no API key

8

Chained agents

Each reads all prior agents' output

7

Deliverables

Architecture to CFO narrative

15

Scenarios

Widest vertical coverage in portfolio

5

JSON repair attempts

Most resilient parsing in the portfolio

0

AI-calculated finance

ROI/NPV computed deterministically

LIVE DEMO · NO API KEY REQUIRED

Try it right now

[Live demo → danvzla.github.io/ai-architecture-advisor](https://danvzla.github.io/ai-architecture-advisor)

[GitHub → github.com/danvzla/ai-architecture-advisor](https://github.com/danvzla/ai-architecture-advisor)



- ◆ Select Demo Mode → Quick Demo → full 8-agent blueprint in ~8 seconds, no key
- ◆ 7 output tabs: Architecture · ADRs · Risk · Roadmap · ROI · CFO Narrative · Scorecard
- ◆ 15 pre-built scenarios spanning Telco, Financial Services, Healthcare, Gov't & more
- ◆ Financial math computed deterministically — never hallucinated by the AI

Daniel Mazzini

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AI · Architecture Advisor